

Project Scout

Master Project Summary
Last updated: June 12, 2026 | Version 25

Upload this document at the start of every new Claude or ChatGPT conversation about Scout. This is the single source of truth.

June 12, 2026 — What Changed Since Version 24

- ✓ Wake word filter **COMPLETE** — Scout only responds when he hears his name.
- ✓ Conversation window added — 30 seconds after Scout responds, no need to repeat his name.
- ✓ Boot window added — Scout is awake and ready immediately after booting. No name needed.
- ✓ 'Connect to the internet' phrase added to ScoutIntentRouter.kt — and several variations.
- ✓ Memory recall bug **FIXED** — TeachExtractor regex was too strict. Removed trailing word boundary. Scout now recalls facts reliably after other questions.
- ✓ Gemini online/offline flow confirmed working — boots offline (TinyLlama), 'go online' activates Gemini.
- Face recognition still pending — Scout sees people but does not know who they are.
- API key still hardcoded — saving removal for Settings screen session.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

- Diana: Patrick's wife
- Elijah: Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
- Nicolas: The family dog. The Nicolas Protocol: Scout stops immediately when the dog is detected.

Names must NEVER be hardcoded in Scout's code. Always use variables.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a calm family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

Item	Detail
Package	com.example.scoutface
Language	Kotlin + C++ NDK
Primary device	Samsung Galaxy A32 — stress-test device, used for all testing via WiFi
Dev device	Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
Future hardware	KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
Ship target	Google Play Store — 7-day free trial, then \$9.99 one-time purchase. No subscriptions. Ever.

3. Identity & Purpose

Scout should feel: Calm. Thoughtful. Emotionally subtle. Grounded. Occasionally curious. Sometimes unsure. Quietly alive. Honest. Predictable. Present.

Scout should NOT: Constantly praise the user. Act overly excited. Feel fake or scripted. Behave unpredictably. Constantly force conversation. Use permanent goodbye language.

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

Trial: 7-day free trial on Google Play. Families get to know Scout, fall in love with him. The \$9.99 feels like nothing because they already care about him.

Purchase: \$9.99 one-time purchase. No automatic charges. No recurring fees. No subscriptions. Ever.

Post-trial strategy: After 7 days, advanced features lock but Scout stays installed. He still shows his face. He still greets the family. Trial end message: 'Thank you for spending time with Scout. Scout is still growing and receiving updates. If you'd like to continue the journey, you can unlock the full version at any time.'

Baseline Brain: TinyLlama 1.1B Chat Q4_K_M (~669 MB) — default, offline, always included.

Optional Gemini: Users add their own free Gemini key in Settings. OFF by default. Scout NEVER ships with a bundled key.

5. Support Scout (In-App, Optional)

Tier	Amount	Label
Coffee	\$3	Buy Scout a Coffee
More	\$5	Support Scout More
Grow	\$10	Help Scout Grow
Founding	\$25	Founding Supporter

Support Scout screen designed and ready. Message: 'You're not just supporting an app — you're supporting a companion.'

6. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done — FLEXIBLE
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

- Sovereign Rules: SQLite Truth always overrules everything.
- Nicolas Protocol: dog detection → immediate stop.
- Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'
- Flexible Memory: Scout stores and recalls ANY fact — not just hardcoded family slots.

7. Current Technical State

Working:

- ✓ Animated face (ScoutFaceView) — mouth, iris drift, thinking expression
- ✓ Speech recognition (Android STT) + Text to Speech (TTS)
- ✓ Camera — face detection (ML Kit), scene labeling
- ✓ Gemini API integration — OFF by default, activated by voice command 'go online'
- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather, IDENTITY, RECALL_FACT
- ✓ Teaching system — flexible: 'my favorite color is teal' → stored permanently
- ✓ Recall system — FIXED June 12. Recalls facts reliably after other questions
- ✓ Wake word filter — Scout only responds when he hears his name
- ✓ Conversation window — 30 seconds after responding, name not required
- ✓ Boot window — Scout awake and ready immediately after boot
- ✓ Weather — current, tonight, tomorrow, specific day, 7-day, precipitation % via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B on BOTH A32 and Fold 7
- ✓ Identity questions hardcoded — routing expanded
- ✓ Total offline mode — 'go offline' blocks ALL internet features
- ✓ Thinking-state expression — drift, narrowed lids, asymmetric brows
- ✓ Online phrases expanded — 'connect to the internet' and variations now recognized

Pending — Launch Blockers:

■ **Face recognition — Scout sees people but does not know who. ML Kit + PeopleDb + TruthDb need connecting.**

■ **Remove hardcoded Gemini API key — line 345 in MainActivity.kt. Saving for Settings screen session.**

- Basic Settings screen — API key entry, offline toggle, robot name display minimum.
- Startup diagnostics — friendly message if brain, TTS, or STT missing at startup.
- Fold 7 stability testing — all testing done on A32 via WiFi. Fold 7 not yet tested.
- Proposal Sandbox — 'Want me to remember that?' confirm step.
- Permanent vs temporary memory sorting.
- Caring follow-up loop.
- ScoutFaceView cleanup — 2 dead-code lines.
- Response cleanup layer — post-TinyLlama filter.
- Scout news feed — FUTURE feature.
- MainActivity cleanup — excessive blank lines throughout file.
- Offline Brain Phase 3 — Google Play Asset Delivery.
- Wire in full mood system.

7b. b8946 API Discoveries — CRITICAL

llama_vocab is a separate type: llama_tokenize and llama_token_to_piece now require const llama_vocab*. Get via llama_model_get_vocab(model).

Functions that crash — hardcode instead: llama_n_vocab → 32000. llama_token_eos → 2. llama_token_eot → 2.

KV Cache: llama_kv_cache_clear() does not exist. Free and recreate context each call.

Logits: After prefill, use n_prompt-1. In generation loop, use index 0.

Backend: ggml_backend_load_all() returns 0 on Android. Use ggml_backend_load(cpuPath) with explicit .so path.

Struct padding: _pad[508] trailing array on params structs. Do NOT remove.

Tuning (A32)	Value
n_threads	2 (memory-bound)
nPredict	64 (~38 tokens, ~7.6s)
History	2 turns
Speed	~15 tok/s prefill, ~4 tok/s generation
Reality	20–40s per answer — acceptable, Gemini is fast path

7c. Known Issues — Do Not Touch Without Discussion

Issue	Notes
Face recognition	Scout sees people but does not know who. Says object labels not names. ML Kit + PeopleDb + TruthDb need connect
Hardcoded API key	Patrick's personal Gemini key in MainActivity.kt line 345. Removing in Settings screen session.
Fold 7 not tested	All testing done on A32 via WiFi. Fold 7 needs a dedicated stability session.
TinyLlama slow on A32	20-40s per answer. Expected. Hardware limitation. Gemini is fast path when online.
Iris jitter (A32 idle)	Hardware timing issue. NOT present on Fold 7.
Barge-in	Deliberately disabled. Runaway loop. Status: PARKED.
STT name recognition	'Scout' misheard as 'Gal', 'Scott', 'Out'. Partially handled by wake word filter.
Live news	Neither brain reads live news. Future news feed needed.
ScoutFaceView dead code	Line 1023: doubled condition. Line 709: unused browAsym. Harmless but messy.
MainActivity blank lines	Excessive blank lines throughout file. CTRL-ALT-L does not remove them. Cosmetic only.

7d. Session Log

- June 5: IDENTITY intent + hardcoded responses. Weather offline fix. Total offline mode.
- June 7: TinyLlama A32 crash stabilized. Identity routing expanded. Face direction locked.
- June 8: Thinking expression — gentle iris drift, narrowed lids, asymmetric brows. Flexible Memory Planning Document created.
- June 9: Flexible memory foundation built. RECALL_FACT intent added. First autonomous memory recall.
- June 10: Business model updated — 7-day free trial. Wake word identified as urgent. Face recognition identified as launch blocker. Launch Checklist created. Spanish language support planned.
- June 12: Wake word filter built. Conversation window added (30 seconds). Boot window added. 'Connect to the internet' phrases added. Memory recall bug fixed — TeachExtractor and handleRecallIntent regex too strict, trailing word boundary removed. Gemini online/offline flow confirmed. All testing on A32 via WiFi.

8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F and CTRL-R approved for large files — always specify which tab first.
- Some Scout files have NO indentation (flush-left). If a search fails, try a shorter unique single-line string.
- Some logic lives in TWO places (e.g. updateLife AND scheduleNextFrame) — change both or Scout flickers.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.

- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear and concise.
- MainActivity.kt has excessive blank lines — CTRL-ALT-L does not fix them. Work around them with line numbers.

9. Key Files

File	Description
MainActivity.kt	Main app — all logic. API key line 345 — REMOVE BEFORE LAUNCH. Wake word filter in onResults(). handling
ScoutFaceView.kt	Custom face canvas — all visual animation. Thinking expression updated June 8.
ScoutIntentRouter.kt	Intent routing — IDENTITY + RECALL_FACT added. Online phrases expanded June 12.
TeachExtractor.kt	Extracts facts from speech — FLEXIBLE. Regex fixed June 12 — trailing \b removed.
FactKey.kt	Fact labels — fixed keys kept + FactKey.custom() for any new label.
TruthDb.kt	SQLite fact store — fully flexible. No changes needed.
ApiKeySetupActivity.kt	API key wizard — built by Claude. Needs wiring to secure storage.
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — precip %, offline-aware
ScoutPresenceDecider.kt	Social timing layer
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
HabitLayer.kt	Pattern memory — 14-day decay
PeopleDb.kt	People memory — needs connecting to face recognition vision response

10. Scout Animation Goal & Mood System

Scout's face should feel alive and emotionally present, but always calm. Never perfectly still.

- CALM — Eyes center, brows neutral, subtle smile (0.15). Scout's default.
- CURIOUS — Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
- HAPPY — Eyes center, both brows lift gently (0.12), smile increases (0.35).
- THINKING — Eyes drift up/around, lids narrow, brows asymmetric. PARTLY BUILT June 8.
- CONCERNED — Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

Version	What Changes
v1.0	Current eyes. Stable irises. Better identity. Better memory. Fix iris jitter.
v1.5	Eyebrows move per mood. Smooth natural blink every 4–8 seconds.

v2.0	Simple upper eyelids + simple lower eyelids. Still cartoon. Still Scout.
v3.0	Full emotion system. Scout at most expressive. Still obviously Scout.

- Keep forever: blue iris, white sclera, cartoon style.
- Never add: tear ducts, skin folds, eyelashes, realistic anatomy.
- Design principle: 'Scout stays Scout. He just gets a little more alive.'

11. TinyLlama Safety Architecture

- Layer 1: Hard-coded safety rules (app level). Always enforced.
- Layer 2: Scout identity rules (system prompt + hardcoded intents).
- Layer 3: TinyLlama generation (last line of defense only).

TinyLlama knows profanity and scary content. Scout must NOT rely on it for safety decisions. Future: response cleanup layer after TinyLlama, before TTS.

12. Device Requirements

	Android	RAM	Storage	Notes
Minimum	13+	4 GB	~3–5 GB	Everything works; responses slower.
Recommended	13+	8 GB+	10 GB+	Faster responses, smoother animations.

13. Brain Upgrade Models

Tier	Size	Model
FREE	669 MB	TinyLlama 1.1B Q4_K_M — baseline, always included
\$2.99	1.79 GB	Phi-2 2.7B Q4_K_M — Microsoft. Solid first upgrade.
\$4.99	2.02 GB	Llama 3.2 3B Q4_K_M — Meta. THE most important upgrade.
\$6.99	~2.4 GB	Phi-4 Mini Q4_K_M — Microsoft (PRIMARY Pack 3).
\$6.99 ALT	2.39 GB	Phi-3.1 Mini 4K Q4_K_M — tested backup Pack 3.
\$9.99	4.92 GB	Llama 3.1 8B Q4_K_M — Meta. Flagship, Fold 7 class.

14. Hardware Direction — Optional

Scout works fully without hardware. KEYESTUDIO Mini Tank Kit V2 (Patrick owns one). Hardware Mode is opt-in. The \$9.99 baseline must work fully on the phone alone.

15. Future Vision

Scout notices patterns in user behavior, generates a suggestion for a new behavior, and asks permission before activating it. Nothing changes without explicit consent. Scout cannot write compiled Kotlin — but CAN generate behavioral scripts, response patterns, routing rules, and habit triggers within the existing safe framework.

16. Language Support — Planned

Phase 1 — Early Spanish Support (No new brain model needed)

- Add a Language setting: English / Español (no flags — just the language name).
- Android STT switches to Spanish when selected.
- Android TTS speaks in Spanish when selected.
- All hardcoded responses translated.

Phase 2 — Full Spanish Support (Long-Term Future)

- Evaluate Spanish-capable offline brain models when brain pack infrastructure is mature.

Priority: Not now. Current roadmap comes first.

17. Play Store Launch Checklist

#	Task	Status
1	Wake word filter	✓ DONE June 12
2	Memory recall bug	✓ DONE June 12
3	Face recognition — connect ML Kit to PeopleDb and TruthDb	Not started
4	Remove hardcoded Gemini API key — saving for Settings session	Not started
5	Basic Settings screen — API key, offline toggle, robot name	Not started
6	Startup diagnostics — friendly message if systems missing	Not started
7	Fold 7 stability testing	Not started
8	A32 stability testing	Ongoing — all June 12 work tested on A32